DASH Azure Blob Access Guide

Prerequisites

1. Azure CLI: Install and configure the Azure CLI on your local machine to run the commands below.
    - **Install Azure CLI**

Input Guide

This outlines the input parameters available during Azure Blob provisioning via the DASH Portal. Each row describes a specific parameter, including its meaning, possible values, recommended values, and any valid combinations.

Input Parameter	Description	Values Provided	Value Guidance	Valid Combinations
Region	Azure region where resources will be deployed.	East US, Central US, West US 2, Central India, UK South, Canada Central, East Asia, South Central US	Choose region closest to workload for better latency.	N/A
Storage Account Tier	Defines the performance tier of the storage account.	Standard, Premium	**Standard:** Default performance tier suitable for general-purpose storage. **Premium:** High-performance tier for BlockBlobStorage or FileStorage.	Premium requires account kind BlockBlobStorage or FileStorage.

Storage Account Redundancy	Specifies how data is replicated across Azure regions.	LRS, GRS, RAGRS, ZRS, GZRS, RAGZRS	**LRS:** Locally Redundant Storage (3 copies within one datacenter) **GRS:** Geo-Redundant Storage (replication to a secondary region) **RAGRS:** Read-Access GRS (GRS + read access to the secondary region) **ZRS:** Zone-Redundant Storage (replication across availability zones) **GZRS**: Geo-Zone Redundant Storage (ZRS + cross-region) **RAGZRS:** Read-Access GZRS (GZRS + read access to secondary)	Premium supports only LRS & ZRS; Standard supports all.

Storage Account Kind	Type of storage account to be created.	StorageV2, BlobStorage, BlockBlobStorage, FileStorage, Storage	**StorageV2:** Recommended. Supports all modern features including Data Lake Gen2 **Blob Storage**: Blob-only, used in legacy scenarios **BlockBlobStorage**: Premium block blob storage (requires Premium tier) **FileStorage:** Premium file storage (requires Premium tier). **Storage:** Legacy storage account. Limited and not recommended	BlockBlobStorage and FileStorage require Premium tier.

Enable Hierarchical Namespace	Enables Azure Data Lake Gen2 features.	true, false	**true**: - Enables file system-like folder structure - Allows file/folder move and rename operations - Optimized for analytics tools like Databricks, HDInsight, Synapse - Required for Azure Data Lake Gen2 features **false**: - Default option - Standard blob storage behavior (no folder hierarchy)	is_hns_enabled = true is recommended when the storage account is intended for big data analytics or when using **Data Lake Gen2** features. Compatible with StorageV2 and account tier = Standard (most common use case). LRS, ZRS, GRS, RAGRS, GZRS, RAGZRS supported
Number of Containers	Defines how many blob containers to create.	1, 2, 3, 4, 5	Choose between 1 to 5.	Must match number of container names.
Container Names	Names of containers to be created.	**Values must be provided by the user as a list, based on the number of containers required** Example: ["logs", "backup", "assets"]	Lowercase, numbers, dashes only.	List size must match number of containers.
Container Access Type	Controls public access to containers.	blob, container, private	**blob:** Blob-only access **container**: Full public access **private**: No access	N/A

owner	The team or individual responsible for provisioning, managing, and maintaining the resource.	Specify the team name or individual responsible for maintaining the Azure blob.	Provide correct team/individual name.	N/A
application	Service or application associated with the blob.	Enter the name of the application or service that uses this blob.	Provide a service name linked to the blob.	N/A
project	Project or initiative name linked to the resource.	Indicate the project or initiative name related to the resource.	Provide the project name related to the resource.	N/A
deployment	The target environment (e.g., development, testing, prod) where the resource will be provisioned.	Define the deployment environment name (e.g., dev, test, prod)	Choose the environment where the Azure blob has to be deployed.	N/A
Services expire after 30 days. Create this service now, or schedule for a later date?	Azure blob will be deleted 30 days after creation, or it can be scheduled to be created at a later date.	Create Now (or) Schedule for Later	Choose based on resource requirements.	N/A

Valid Tips:

Storage Type	Tier	LRS	ZRS	GRS	RAGRS	GZRS	RAGZRS
StorageV2	Standard	✅	✅	✅	✅	✅	✅
BlobStorage	Standard	✅	✅	✅	✅	❌	❌
BlockBlobStorage	Premium	✅	✅*	❌	❌	❌	❌
FileStorage	Premium	✅	✅*	❌	❌	❌	❌
Storage (v1)	Standard	✅	✅	✅	✅	❌	❌

Sample Input Selection:

REGION	East US
STORAGE ACCOUNT TIER FOR EXTERNAL LOCATION	Standard
STORAGE ACCOUNT REDUNDACY FOR EXTERNAL LOCATION	LRS
STORAGE ACCOUNT KIND FOR EXTERNAL LOCATION	StorageV2
ENABLE HIERARCHICAL NAMESPACE	false
NUMBER OF CONTAINERS	2
ENTER CONTAINER NAMES	["red1","blue1"]
CONTAINER ACCESS TYPE FOR EXTERNAL LOCATION	blob
OWNER	dashuser
APPLICATION	dashapp
PROJECT	dash
DEPLOYMENT	dev

Services expire after 30 days. Create this service now, or schedule for a later date?

(●) Create Now ( ) Schedule for Later

ADD SERVICE CANCEL

Output Guide

This document outlines the output values generated after Azure Blob Storage provisioning through the DASH Portal. These values can be used to access, manage, and interact with your storage containers.

Outputs	Description	Example Value
storage_account_name	The name of the Azure Storage Account that was provisioned.	dashstg1zsv
Storage_account_primary_access_key	Primary access key used to authenticate with the Storage Account.	z0Q6Yw9i7eF8fYABJqzRE+abcd12345abcdefg== (truncated)
storage_container_names	A list of Azure Blob container names created within the storage account.	["logs", "backup", "assets"]
list_storage_containers_command	Azure CLI command to list all containers using name and access key. az storage container list --account-name <storage_account_name> --account-key <storage_account_primary_access_key> --output table	az storage container list --account-name dashstg1zsv --account-key z0Q6Yw9i7eF8fYABJqzRE+abcd12345abcdefg== --output table

What You Can Do with These Outputs

Use the following Azure CLI commands to perform various operations on your blob containers. Replace placeholder values like <your-file>, <blob-name>, etc., with actual values.

Operation	Description	Command	Example Command
Upload a Blob	Uploads a file from local machine to a container in Azure Blob Storage.	az storage blob upload --account-name <name> --account-key <key> --container-name <container> --name <blob_name> --file <file_path>	az storage blob upload --account-name ***dashstg1zsv*** --account-key ***z0Q6Yw9i7eF8fYABJqzRE+abcd12345abcdefg==*** --container-name ***logs*** --name ***red1*** --file ***./myfolder/myfile.txt***
List Blobs	Lists all blobs stored in a specific container.	az storage blob list --account-name <name> --account-key <key> --container-name <container> --output table	az storage blob list --account-name ***dashstg1zsv*** --account-key ***z0Q6Yw9i7eF8fYABJqzRE+abcd12345abcdefg==*** --container-name ***logs*** --output table
List Blobs with URLs	Lists blobs in a container along with accessible URLs.	az storage blob list --account-name ***<storage_account_name>*** --account-key ***< storage_account_primary_access_key >*** --container-name ***<container_name>*** --query '[].{name:name, url:join(``, [`https://***<storage_account_name>***.blob.core.windows.net/***<container_name>***/`, name])}' --output table	az storage blob list --account-name ***dashstg1zsv*** --account-key ***z0Q6Yw9i7eF8fYABJqzRE+abcd12345abcdefg==*** --container-name logs --query '[].{name:name, url:join(``, [`https://***dashstg1zsv***.blob.core.windows.net/***logs***/`, name])}' --output table

Download a Blob	Downloads a specific blob from a container to local machine.	az storage blob download --account-name ***<storage_account_name>*** --account-key ***< storage_account_primary_access_key >*** --container-name ***<container_name>*** --name ***<Replace with blob name>*** --file ***<Replace with destination path>***	az storage blob download --account-name ***dashstg1zsv*** --account-key ***z0Q6Yw9i7eF8fYABJqzRE+abcd12345abcdefg==*** --container-name ***logs*** --name ***red1*** --file ***./downloads/myfile.txt***
Get a Blob URL	Generates the public URL for a blob in the container.	az storage blob url --account-name ***<storage_account_name>*** --account-key ***<key>*** --container-name ***<container_name>*** --name ***<Replace with blob name>***	az storage blob url --account-name ***dashstg1zsv*** --account-key ***z0Q6Yw9i7eF8fYABJqzRE+abcd12345abcdefg==*** --container-name ***logs --***name ***red1***